STI Portal / ATIO KB — Requirements Register

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1 Purpose & scope

This register states **what** the redesigned STI Portal / ATIO KB must do and which qualities it must have. It deliberately excludes:

- business rationale and funding logic → CAPEX proposal,
- architecture and technology choices (joinup, CQRS dual-DB, triplestore, Solr, Milvus, the JSON import schema) → Design Doc v4.2 and Architecture Decision Records (ADRs).

Each requirement referencing a design choice points to the ADR that owns it, so requirements stay stable while the ”how” evolves under agile.

2 Glossary

AFSI Agrifood System Innovation — the **content** / knowledge unit. This encompasses both original records and ATIO innovation records or other representations in the platform.

AFSA Agrifood System Actor — a **person or organisation** (policymaker, innovator, farmer rep, researcher, funder).

Claim an AAA-style statement a collection makes about an AFSI; stored alongside slim core fields rather than forced into a fixed schema.

Collection a data source’s own body of records, retaining its own vocabulary/taxonomy.

Connected a notebook-style entry of a AFSA in another AFSA’s profile

Relevant (to an actor) Topic or interest matches AFSA’s interests

AAA ”Anyone can say Anything about Anything” — the semantic-web principle underlying the claims model.

FAIR Findable, Accessible, Interoperable, Reusable.

SKOS Simple Knowledge Organization System (vocabulary/taxonomy exchange).

RAG Retrieval-Augmented Generation — an application pattern over a vector-search endpoint, not itself a query interface.

ADR Architecture Decision Record.

3 Constraints & assumptions

These are *givens*, not choices. They are not requirements and must not be confused with them.

3.1 PROPOSED Drupal engine is retained

The Drupal engine remains the content-handling and orchestration layer (team skills, consultant pool, de-facto public-sector standard).

3.2 PROPOSED FAO GCP hosting

The system is hosted on FAO Google Cloud Platform, with edge → review → production CD.

3.3 PROPOSED No regression of ATIO curation

The redesign shall not remove any curation capability the ATIO team currently relies on, and existing ATIO curation workflows (analysis, synthesis, standardisation) shall remain operational. (This absorbs the old "existing workflows keep working" acceptance line, which was a non-regression check rather than a forward requirement.)

3.4 PROPOSED Interoperability with FAO infrastructure

The system must interoperate with existing FAO infrastructure and vocabularies (AGROVOC, FAOSTAT, …).

3.5 PROPOSED Deliver the promised scope

The redesign must deliver what the CAPEX proposal and ATIO Design Doc outline.

4 Functional requirements

4.1 Data agglomeration AGG

4.1.1 PROTOTYPE Import preserves original key–value pairs

The system shall import a source record supplied in a supported format, preserving the source's original field names and values without mapping them onto a fixed schema.

- Acceptance: two structurally different sources load through the same path, and every original key–value pair is retrievable afterwards.
- Supported formats are structured data (JSON, XLSX, CSV, XML, SQL).

4.2 Curation CUR

4.2.1 PROPOSED Non-developer curator can author records via GUI

A content curator without software-development skills shall be able to create, edit, and submit an innovation record for review through a graphical interface, without writing code or running CLI commands.

- Acceptance: a non-developer completes record creation → submission in a usability test with no CLI use.

4.2.2 PROTOTYPE Maintainer can curate SKOS relationships

A data curator can match semantic dimensions across collections using SKOS (e.g. 'Technological Readiness' of collection A similar to 'Readiness' of collection B)

4.3 Access & interoperability API

4.3.1 PROTOTYPE SPARQL endpoint per collection

Each data collection shall be queryable by external machine clients via a SPARQL endpoint, returning records annotated with that collection's own vocabulary.

4.3.2 PROTOTYPE REST API per collection

Each data collection shall be queryable by external machine clients via a REST API, returning records annotated with that collection's own vocabulary.

4.3.3 PROPOSED Vector / semantic-search endpoint per collection

The platform shall expose a vector/semantic-search endpoint over each collection, suitable for consumption by RAG clients.

4.3.4 PROTOTYPE Collection-level landing page

The platform shall present one landing page per collection listing this collections AFSIs and filtered search based on that collection's taxonomy.

4.3.5 PROTOTYPE RDF Data Model

The platform shall employ a data model that matches both Unit 1 and Unit 5 requirements.

4.4 Networking NET

Networking depends on solid implementation of the core STI portal (AGG, CUR, API).

4.4.1 PROPOSED Actors linkable to innovations and to one another

Actors (AFSA) should be linkable to the innovations (AFSI) held in central storage, and to one another.

4.4.2 PROPOSED Actor-to-actor messaging

An actor shall be able to send a message to another actor through the platform.

4.4.3 PROTOTYPE Groups / communities of practice

Actors shall be able to form and join groups / communities of practice on the platform.

4.4.4 PROPOSED Surface relevant innovations to an actor

An actor should be surfaced innovations relevant to them.

4.4.5 PROPOSED Surface relevant actors to an actor

An actor should be surfaced other actors relevant to them.

4.4.6 PROTOTYPE Challenge organization

An actor (AFSA) with an organiser role should be able to create an innovation challenge on the platform and collect idea submissions from other actors.

- Out of scope: assessment, scoring, or ranking of submissions (collection only, no adjudication).
- Acceptance: an organiser creates a challenge; a second actor submits an idea to it; the submission is stored and visible to the organiser; no scoring/ranking UI is present.

4.5 Indicators IND

4.5.1 PROPOSED Display external indicators about an innovation

The platform shall display ASTI indicators about a country drawn from external sources (e.g. FAOSTAT).

4.5.2 PROPOSED User-defined assessment criteria

Data owners and platform users may define their own assessment criteria for innovations.

5 Non-functional / quality requirements

5.1 PROPOSED Interoperability & FAIR

[TBD: which standards, which vocabularies, which FAIR criteria are testable.]

5.2 PROPOSED Provenance & AI-origin auditability

[TBD: every field value traceable to its source/claim; AI-generated values flagged and attributable. Make the AAA + AI-origin principles testable here.]

5.3 PROPOSED High-scale ingestion performance

[TBD: throughput / volume target for bulk import.]

5.4 PROPOSED Collection-level access control

[TBD: per-collection roles and transitions.]

5.5 PROPOSED Maintainability

[TBD: configuration-as-code, environment parity.]

6 Traceability

Requirement	CAPEX / Design Doc	ADR	Test
FR-AGG-01	AAA / claims model	ADR-TBD	
FR-CUR-01	problem statement	ADR-007	
FR-API-01	Feature 8	ADR-TBD	
FR-API-02	Feature 8	ADR-TBD	
FR-API-03	Feature 8 / Milvus	ADR-TBD	